



ROBOT DESIGN MVP

SOCIAL ROBOT CO-DESIGN CANVASES

It's time to design your robot MVP (Minimum Viable Product)! Remember the guidelines you defined.

Where and when

What place?
What time of day?
Does the place or time change?

ED triage desk, Mercer General Hospital; used 24/7, every shift.

Robot's role

Is the robot a friend? Teacher? Helper?
Something else?

Clinical decision-support assistant; nurse retains final authority.

Draw a picture

What does the robot look like?
Is it attached to something?
Does it move around?
Can its appearance be modified?

	☐ _____
	☐ _____
	☐ _____
	☐ _____

No physical form - web-based dashboard on ED desktop/tablet screens.

Personality

Does the robot have specific characteristics?
Does it have emotional states, or needs?

No personality or emotions - a neutral, professional clinical tool with a consistent, calm tone.

Context-based behaviour

What external and environmental factors affect behaviour?
What data is used to adapt to context?

Adapts to live vitals, EHR data, symptoms and demographics; flags missing or invalid input.

Connection to systems

Is the robot connected to external systems, such as software, databases, or other robots?
How does it use these systems?

Connects to Mercer EHR, registration and vital-sign monitors; logs all activity.

Interaction modalities

What modalities are inputs to the robot? What modalities does the robot output?

INPUT

☐ voice	☐ movement	☐ screens
☐ sounds	☒ touch	☐ lights
☐ gestures	☐ smell	☒ other <u>keyboard & mouse</u>
	☐ facial expressions	

OUTPUT

☐ voice	☐ movement	☒ screens
☐ sounds	☐ touch	☐ lights
☐ gestures	☐ smell	☐ other _____
	☐ facial expressions	

Interaction flow

Describe the most important interaction of the robot.

Note: only fill the bottom row if your robot is teleoperated.

	BEFORE	DURING	AFTER
USER	Nurse logs in and opens the patient queue.	Nurse selects a patient, reviews vitals/EHR data and the AI-recommended ESI level, confidence score and key factors; accepts or overrides with a reason.	Nurse confirms the triage level for the patient record.
ROBOT	Dashboard retrieves patient demographics and EHR data.	AI analyses vitals and EHR data and displays the predicted ESI level, confidence score and contributing clinical factors in real time.	System timestamps the decision, updates the queue and logs the audit trail.
ROBOT OPERATOR (optional)			



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THRI paper:

